

Ahmed AbdelHady Mohamed Esmail

AI & Machine Learning Engineer

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Results-driven AI & Machine Learning Engineer with a solid foundation in Computer Vision, NLP, and Deep Learning architectures. Proven ability to architect end-to-end AI pipelines, from training custom models to deploying production-ready RESTful APIs using FastAPI and Docker. Adept at bridging traditional Machine Learning with modern Generative AI, leveraging tools like LangChain and LangGraph to build stateful, multi-agent workflows. Passionate about solving complex business challenges through scalable, data-driven solutions and continuous technical growth.

Education

Modern Academy For Computer Science And Information Technology

Bachelor of Computer Science | Graduation Year: 2024

- Grade: 3.18 / 4.00

Work Experience

AI and Machine Learning Intern • AMIT Learning

June 2023 - February 2024 | Cornish El Maadi, Al Maadi, Cairo, EG

- Collaborated with senior engineers to develop and deploy Machine Learning and Deep Learning models using Python and TensorFlow.
- Gained hands-on experience in Computer Vision, NLP, and MLOps, actively contributing to solving real-world data challenges.
- Participated in the full lifecycle of AI projects, from data preprocessing to model optimization and deployment.

Military Service — Completed | Egyptian Armed Forces

February 2025 - March 2026

- Honorably discharged after completing mandatory military service, demonstrating discipline, adaptability, and teamwork in high-pressure environments.

Projects

• Psoriasis Disease Classification System

- Engineered and containerized** a production-ready RESTful API using **FastAPI** and **Docker** to serve a hybrid CNN/ViT computer vision model for dermatological diagnosis.
- Optimized model inference** by converting legacy Keras models to **TensorFlow Lite (TFLite)**, reducing memory footprint and ensuring cross-platform compatibility.
- Implemented robust file-handling pipelines** (asynchronous uploads, temporary file management) to handle concurrent image-processing requests efficiently.
- Tech Stack:** Python, FastAPI, TensorFlow Lite (TFLite), Docker, OOP.

• Omni Enterprise AI Agent (Custom Agentic Workflow)

- Architected a stateful AI agent** using a custom manual ReAct loop (LangChain Core) and in-memory session management, bypassing deprecated abstractions for deterministic, context-aware function calling.
- Developed secure, custom tools** interfacing with mock enterprise databases (JSON) to handle real-time order tracking and HR inquiries without external dependencies.
- Deployed the production-ready workflow** via FastAPI with strict Pydantic validation, leveraging Pure Python and Ollama (Qwen 2.5) for seamless, scalable backend integration.
- Tech Stack:** Python, FastAPI, LangChain Core, Ollama (Qwen 2.5), Pydantic.

• Smart Document Q&A System (Local RAG Engine)

- **Architected and developed** a privacy-first Retrieval-Augmented Generation (RAG) pipeline using **FastAPI**, **ChromaDB**, and **Ollama** to enable semantic search over proprietary PDF documents without data leakage.
- **Implemented document ingestion** utilizing PyPDF and LangChain text splitters to chunk and embed large datasets, optimizing context window usage for LLMs.
- **Engineered robust system prompts** with strict guardrails to eliminate LLM hallucinations and ensure data privacy by running models entirely on-premises (Local).
- **Tech Stack:** Python, FastAPI, ChromaDB, Ollama (Llama 3 / Qwen), LangChain, Nomic-Embed-Text.

• Enterprise AI Automation Suite (n8n Multi-Agent Platform)

- **Architected 19+ production-grade agentic workflows** using n8n to automate complex business operations across Sales, HR, IT Support, RAG System, and Executive Management.
- **Developed a Multi-Agent WhatsApp/Telegram System** with a central router coordinating 5 specialized sub-agents to process text, voice, and image inputs via Airtable.
- **Built an Agentic RAG pipeline** monitoring Google Drive to chunk and store PDFs in Pinecone using Namespace isolation, alongside structured output parsers for strict JSON generation.
- **Tech Stack:** n8n, Docker, LangChain, OpenRouter, Gemini, Pinecone, Ollama, Airtable, Google APIs, OpenAI.

• Applied Deep Learning & Academic Projects

- **Image Caption Generation:** Developed an automated image captioning model achieving 98% accuracy using CNNs for feature extraction and LSTMs/Transformers for sequence generation.
- **Breast Cancer Detection:** Built a robust early detection system using Computer Vision and CNN architectures with advanced data preprocessing.
- **Text Summarization:** Implemented a summarization tool utilizing advanced transformer models (T5) to condense lengthy documents efficiently.
- **Music Recommendation System:** Designed an NLP-based recommendation engine using TF-IDF vectorization and Cosine Similarity, integrated with Spotify API via Streamlit.

Certificates

- **Deep Learning Applications for Computer Vision** | University of Colorado Boulder — Mar 2024
- **Deep Learning Fundamentals** | DeepLearning.AI — Jan 2024
- **Introduction to Computer Vision and Image Processing** | IBM — Jan 2024
- **Associate Data Scientist in Python** | DataCamp — Dec 2023
- **Data Analyst with Python** | DataCamp — Dec 2023

Skills

- **Languages:** Python, JavaScript, SQL
- **AI & Machine Learning:** Deep Learning, Computer Vision, NLP, Generative AI, RAG, AI Agents, Prompt Engineering.
- **Libraries & Frameworks:** TensorFlow, PyTorch, Keras, OpenCV, LangChain, LangGraph, Pandas, NumPy, Scikit-Learn.
- **Backend & DevOps:** MLOps, Model Optimization, FastAPI, Docker, Pydantic, RESTful APIs, Git.
- **Databases & Tools:** PGVector, ChromaDB, Pinecone, Ollama, n8n, PostgreSQL

Languages

- English — Intermediate
- Arabic — Native or Bilingual